

Secondary Storage

Whether you need to back up important files or transport data from one computer to another, a wide array of external storage options are available to you. From a few megabytes of data to several gigabytes, you can be sure to find a unit that suits your storage requirements, and at the same time fits your budget.

Spinning the Disc

CD-ROM drives offer data transfer speeds ranging from 2.5 MBps to 4.7 MBps for the IDE models. SCSI CD-ROM drives are available, but are not something that most mainstream users will consider.

The differences in transfer rates between CD and DVD-ROM drives are very significant. DVD drives are usually adopted by users who have easy access to DVD movies.

CD-R and CD-RWs are the choice of today for inexpensive storage options, and cost between Rs 3,000 and Rs 7,000 for most 48X to 52X drives.

Technology Insight

A CD-ROM drive consists of a laser that is mounted on an assembly that can align it with any part of the disc as it spins. This laser reads the data which is stored in the form of a pit (bit 0) and land (bit 1). DVD-ROM drives differ only by the higher density of information stored on the media.

CD-Writers are similar to regular CD-ROM drives except that they let you burn the data on blank media. The media has a compound layer which, while in a crystalline state, allows the laser beam to pass through and reflects it back from the metal layer. When in the amorphous state it doesn't allow reflection of light. This describes the 1 and 0 states of information storage. The laser beam is powerful enough to change the state of the dye in the medium used for writing.

Smart Tips

Capacity: This really depends on your application, or the quantity of data you need to back up. There is no real need to go overboard on storage capacity, as the cost of such media and devices are quite steep.

Speed: Random access media such as hard drives, CD flavours

and cartridges have the advantage of quick access to data stored on the media. If you have larger data backup requirements then opt for a drive with higher speed to save time.

Portability: If you share your data with others, it makes sense to consider options that are in widespread use. Iomega's Zip drives, for example, are quite popular. The other options are the newer Flash-memory-based drives that utilise the USB port on any PC and make data portability even simpler.

Build quality: Some cheaper drives have very fragile loading and unloading trays. This tends to fail over a period of time. Look for a drive with a sturdy mechanism.

Durability: Media that is enclosed in a case will withstand the test of time better than exposed media. CD-ROM discs, which are the most popular means of data storage nowadays, are highly susceptible to scratches—which could render them useless.

Interface: IDE is the least expensive and most popular option. SCSI models require an additional SCSI card to be installed, but result in lower CPU utilisation. External models tend to use the USB interface.



Jargon Buster

Back-up: The essential job of saving and storing data on a tape or other storage device, such as a CD-ROM, outside your computer hard disk or computer network.

Copy Protection: Both discs and players come equipped with hardware mechanisms to prevent the illegal copying of DVDs. You will not be able to copy the movie to a VHS tape, your computer's hard drive or any other data-recording device.

DAT: Digital Audio Tape (DAT) is a compact, high capacity form of data storage, suitable for archiving or backing up large amounts of data.

DVD: DVD (digital versatile disk) is a high-density format for playing full motion video. It provides vast data storage capacity. Although popular within the home market, DVDs can also be used to hold computer data.

Decision Maker

	Inexpensive backup	Fast backup	Portability and space	Entertainment enthusiast
You Need	Speed is not an issue; you are just looking for a cheap, reliable backup solution	Speed is everything, and you have no intention of spending more than 5 or 6 minutes to burn a CD.	Speed becomes a concern as well. Price is not a problem.	To watch high quality movies as well as have the regular CD-ROM functionality
Look For	32X, 48X CD-ROM	A 52X CD-Writer	A drive that can store approximately 1 to 2 GB of data on a single disc; SCSI or FireWire connectivity.	Look for a DVD-ROM that is rated at 16X or better.
Our Pick	Samsung, LG or Liteon	Liteon LTR55246S, MSI CD-RW CR52-A2	Iomega Jaz drive, Castlewood ORB drives	LITEON 16X DVD Drive, Pioneer 16X DVD Drive
Price Range	Rs 3,000 to 3,500	Less than Rs 5,500	Rs 17,000 to Rs 28,000	Rs 2,000 to Rs 5,000